
A Small Internal Signal Marks Thinking-Mode Entry in Open Language Models

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Abstract

1 Some open language models let users turn “thinking” on or off. We ask whether
2 the activation immediately before generation records that choice and causally se-
3 lects the response branch. For three models, a model-specific thinking-minus-off
4 vector averaged over the last two tokens of 60 training prompts moves held-out
5 distributions toward thinking in Qwen3.5-4B, Gemma 4 E2B-it, and Nemotron
6 Nano 4B; exact copies localize the effect to response entry. With Qwen3-4B’s soft
7 switch enabled in both modes, copying one shared branch-token activation selects
8 the donor token on 24/24 localization tests. Fixed controllers of widths 1, 2, 4, and
9 8 each do so on 72/72 held-out tests (width-1 controls: 0/432), and one copied to-
10 ken determines donor response format on 72/72 continuations across three seeds.
11 On 30 held-out non-math prompts and ten seeds, one Nemotron vector decision
12 yields an exact thinking opening on 293/300 tests versus 0/300 for random vectors.
13 Under a protocol frozen before any AIME 2026 generation, the same intervention
14 enters thinking on 30/30 problems and scores 4/30 strictly correct, versus 1/30
15 for off and random and 7/30 for canonical thinking; each paired exact test has
16 $p = .25$. It reaches the correct numerical result on all seven problems scored
17 correct by thinking, but fails to stop on three; only 5/30 fixed responses finish
18 under the shared 8,192-token cap. Repeated intervention corrupts text. The activa-
19 tion causally selects response entry, but does not establish reasoning recovery or
20 full-answer control.

21 1 Introduction

22 Open language models increasingly expose a “thinking” button, a template option, or a system
23 instruction that changes how the model answers. These controls affect latency, cost, response length,
24 and answer style. But a visible <think> tag or a long response does not by itself show what changed
25 inside the model.

26 Three things must be kept separate:

- 27 1. the visible text or setting that selects a mode,
- 28 2. the internal activation signal present just before generation, and
- 29 3. the later response, including its length, style, and accuracy.

30 This paper studies the second item. We ask whether the mode choice leaves a measurable internal
31 signal at the end of the prompt and whether changing that signal changes which response style starts.
32 We do not claim that this signal is reasoning itself.

33 The basic test is simple. For the same user prompt, run the model once with thinking on and once
34 with it off. At the last two prompt tokens, subtract the off-mode activations from the on-mode

35 activations. Average that difference over public training prompts to obtain one model-specific *mode*
 36 *vector*, denoted c_m . Freeze the vector, add it to held-out off-mode activations, and check whether the
 37 model’s next-token probabilities move toward those of the held-out thinking run. The intervention
 38 sees the held-out off-mode state and the training vector only; the held-out thinking activation is used
 39 only afterward for scoring. We call this a *prompt-end mode-signal audit*.

40 This question is related to token-level reasoning triggers. Mid-Think shows that visible strings such
 41 as Okay, </think>, and newline patterns can change response style (Yang et al., 2026). Here the
 42 visible switch is held fixed and audited: we directly modify the activation signal it produced. The
 43 main result is that several public switches write a small, causal signal at the end of the prompt. The
 44 important limitation is equally simple: entering a response mode is not the same as producing a
 45 correct or stable long response.

46 **Contributions.** This paper makes five contributions.

- 47 • We introduce a held-out activation test for whether a public thinking switch writes a recov-
 48 erable mode signal at the end of the prompt.
- 49 • We show that fixed, model-specific mode vectors learned from 60 public paired prompts
 50 transfer to held-out prompts in Qwen3.5-4B, Gemma 4 E2B-it, and Nemotron Nano 4B.
- 51 • We localize the causal signal to the last prompt tokens in Qwen3.5 and Nemotron. In
 52 Qwen3-4B, we use the public soft switch and localize causal control to the shared <think>
 53 state immediately before the modes diverge. Controllers learned on 12 prompts transfer on
 54 36 further prompts; a width sweep shows that the one shared position is sufficient. Copying
 55 the full post-attention residual state reproduces the donor effect; copying only the MLP
 56 update does not reproduce the full effect.
- 57 • We test simpler explanations using reversed and random vectors, earlier-token copies, al-
 58 tered suffixes, masked opening tokens, and misleading user instructions.
- 59 • We replicate Qwen3’s selected branch token on 12 disjoint, category- and template-matched
 60 prompts with three sampling seeds; test Nemotron generation on held-out non-math
 61 prompts and exact copies on six math prompts; and run a paired one-step answer-quality
 62 experiment on all 30 AIME 2026 problems under a protocol frozen before any AIME 2026
 63 generation. We release row-level results, a manual trace audit, and an independent recom-
 64 putation audit.

Table 1: Audit overview. Each row gives one component of the evidence chain.

Audit question	Main evidence	Result signal
Does a public switch leave a reusable internal signal? Where is the signal?	A fixed vector learned on public prompts is added to held-out off-mode activations. Copy the final prompt activations, or Qwen3’s shared branch token; compare with earlier positions.	Next-token probabilities move toward the held-out thinking run. Response-entry copies work; matched earlier copies do not.
Which intermediate state is sufficient?	Copy the full post-attention residual state, or copy only the MLP update.	The former reproduces the donor effect; the latter does not reproduce it fully. This asymmetric test does not isolate the attention update.
Is this only suffix or first-word copying?	Alter suffixes, mask likely opening tokens, reverse the vector, use random vectors, and contradict the mode in user text.	The activation effect survives these controls.
What answer quality follows a successful entry?	Grade AIME 2026 after one fixed-vector entry; also test 64 repeated fixed-vector or exact-copy steps.	Fixed entry is 4/30 correct versus 1/30 off/random and 7/30 thinking, with $p = .25$ and only 5/30 completions; repeated intervention can corrupt text.

65 Table 2 separates the four same-checkpoint tests from secondary paired-checkpoint comparisons.
 66 This distinction matters: only the same-checkpoint rows study an inference-time switch.

67 2 Setup

68 **Modes and models.** The main fixed-vector tests use three open models with public thinking con-
 69 trols: Qwen3.5-4B (Qwen Team, 2026), Gemma 4 E2B-it (Google DeepMind, 2026), and Llama-
 70 3.1 Nemotron Nano 4B v1.1 (NVIDIA, 2025). The new causal audit uses Qwen3-4B (Qwen Team,

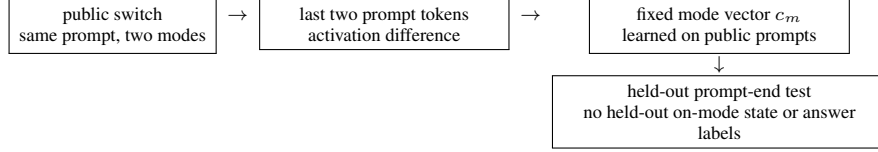


Figure 1: Main protocol. Public paired prompts produce a fixed mode vector. On a held-out prompt, the vector is added to the off-mode activation before the first generated token. The corresponding held-out on-mode activation is not used by the intervention; it is used only to score the resulting probability distribution.

Table 2: What changes between the two modes, what we test, and what we do not infer. “Exact copy” means copying the observed activation from one run into the other; “fixed vector” means adding one average difference learned on other prompts.

Model	How mode is selected	Evidence in this paper	Important limit
Qwen3.5-4B	Same checkpoint; chat-template suffix	Exact copying, fixed vector, broad and contest prompts, controls	One fixed vector is weaker on its 30 broad prompts.
Gemma 4 E2B-it	Same checkpoint; template switch	Fixed vector on broad and contest prompts	Reverse fixed-vector direction is unreliable.
Nemotron Nano 4B	Same checkpoint; system prompt	Exact copying, fixed vector, broad prompts, one-step and sustained generation, AIME 2026 answer quality	Fixed entry reaches 4/30 correct and 5/30 complete; canonical thinking reaches 7/30 correct. Repeated intervention corrupts text.
Qwen3-4B	Same checkpoint; <code>enable_thinking=True</code> with <code>/think</code> or <code>/no_think</code>	Exact-copy localization; width 1/2/4/8 fixed controllers on 36 held-out prompts; one-token follow-through	The fixed controllers test response entry, not answer accuracy or continued control.
OLMo 3 7B	Separate Think and Instruct checkpoints	Exact copying and fixed-vector comparison	Not an inference-time same-checkpoint switch.
Phi-4 mini / Liquid 1.2B	Separate reasoning or thinking and instruct checkpoints	Secondary exact-copy localization	Not an inference-time same-checkpoint switch; Liquid is asymmetric.

2025). All four compare two modes of the same checkpoint. OLMo 3 7B (OLMo Team, 2026) is analyzed separately because it compares Think and Instruct checkpoints. Earlier Phi and Liquid experiments remain secondary appendix results. In the audited Qwen3.5 template, the assistant suffix differs as follows:

```

think      <|im_start|>assistant\n<think>\n
no-think   <|im_start|>assistant\n<think>\n
              \n</think>\n\n

```

The shared prefix is intentional: in this Qwen3.5 template, no-think opens and immediately closes an empty think block before the answer begins.

Qwen3-4B uses its documented soft switch instead. Both conditions keep `enable_thinking=True`; the user message ends in `/think` or `/no_think`. The two prompts then share the same assistant header, and both modes first emit `<think>`. Their next-token distributions sharply separate: thinking mode’s top token is the single-newline continuation that begins nonempty reasoning, while no-think’s top token is the double-newline continuation of an empty block. We therefore supply the same `<think>` token to both conditions and perform the Qwen3 causal audit at that first divergent decision. This is not the hard template switch obtained with `enable_thinking=False`.

Prompt sets. Table 3 states exactly where prompts come from and how they are used. The fixed vector never sees an AIME prompt during fitting or selection.

Terms used below. An *activation* is the vector of numbers passed between transformer layers. The *residual stream* is this main layer-to-layer path. A *donor* is the run whose mode we copy; the *target* is the run we modify. A *tail patch of width 2* copies activations only at the final two audited tokens. A *fixed mode vector* is one average activation difference learned on other prompts. Appendix G gives intuitive and mathematical definitions together.

Table 3: Prompt sources and roles. Selection is deterministic; these are not random samples of all possible prompts.

Role	Source and selection	Use
Fixed-vector training	First 30 usable GSM8K test questions and first 30 MATH-500 test rows with a simple numeric final answer	Learn the mean activation difference and choose its scale.
Held-out contest test	All 30 AIME 2025 and all 30 AIME 2026 questions	Prompt text only; no answers, correctness labels, or thinking-mode activations enter the intervention.
AIME 2026 answer-quality test	All 30 AIME 2026 questions; protocol frozen after controller fitting and before any AIME 2026 generation	Grade the ordinary continuation after one patched entry decision; gold answers are used only after generation.
Broad transfer	Authored instruction, coding, factual, multi-turn, and non-English prompts; Nemotron also uses GPQA Diamond	Test transfer outside the math training distribution.
Localization and controls	Fixed, category-balanced authored suites; 12 Qwen3-4B localization prompts	Locate the effect and compare post-attention state copies with isolated MLP-update copies.
Qwen3 prompt-value replication	The next two authored instances in each of the same six categories and templates; 12 prompts disjoint from localization	Test whether the selected branch token determines response format under three sampling seeds.
Qwen3 fixed-controller audit	The final two instances in those six categories, plus all 24 prompts from four categories absent from fitting; 36 prompts disjoint from fitting and the prompt-value replication	Test frozen signed controllers of widths 1, 2, 4, and 8 at the shared <think> branch point.

93 The layer scans use different grids. Qwen3.5 tests every residual-stream boundary from 0 through 32.
 94 Qwen3 tests boundaries 0, 4, . . . , 32, and 36; Nemotron and OLMo test 0, 4, . . . , 32. Boundary 0 is
 95 the embedding output passed into the first transformer block. Tables report the first tested boundary
 96 where every prompt moves closer to the donor. For Qwen3.5 this is the earliest boundary in a dense
 97 scan. For the other models it is only the first successful boundary on a four-layer grid, not an exact
 98 onset or a unique causal source.

99 Throughout the paper, intervention labels are written as *donor mode into target mode*. For example,
 100 “think into no-think” means that activations are copied from a thinking run and inserted into the
 101 corresponding no-think run before comparing next-token probabilities.

102 **Patch metric.** Let p_D and p_T be the donor and target final next-token distributions for the same
 103 user prompt under two modes, and let p_P be the distribution after patching target residual state with
 104 donor information. We report donor win when

$$\text{JS}(p_P, p_D) < \text{JS}(p_P, p_T),$$

105 where JS is Jensen–Shannon divergence, a bounded measure of how different two probability dis-
 106 tributions are. A donor win therefore means that the modified run is closer to the intended mode
 107 than to its unmodified mode. We also report margin retention for low-rank patches: the fraction of
 108 the exact-copy improvement retained by the smaller patch. Confidence intervals resample the stored
 109 prompts. They do not cover new checkpoints or all possible prompts, and exact rates such as 60/60
 110 are finite-suite results rather than population guarantees.

111 **Fixed mode vector.** The main reusable intervention is one model-specific vector, c_m , that moves
 112 a prompt-end activation toward thinking mode:

$$h_{\text{tail}} \leftarrow h_{\text{tail}} + \alpha c_m.$$

113 For the three original fixed-vector models, c_m is the average thinking-minus-off difference at the
 114 final two prompt tokens over 60 training prompts. For Qwen3-4B, separate width-1/2/4/8 mean
 115 residual tensors are learned from 12 prompts at boundary 28, added for off-to-think, and subtracted
 116 for think-to-off. Each scale is selected on those 12 prompts before evaluating the same 36 held-out
 117 prompts. A held-out test uses only the target activation and frozen tensor; the donor activation is
 118 used afterward for scoring.

119 **Exact activation copies.** For a layer boundary ℓ and width w , an exact tail patch replaces the
 120 target activation at the final w prompt positions with the activation observed in the donor run. This is
 121 stronger than the fixed-vector test because it uses the matching donor run. Its purpose is localization:
 122 matched controls copy the same number of earlier tokens. A separate asymmetric test copies either
 123 the full residual state after attention or only the MLP update.

124 **Low-rank residual code.** For each prompt i , direction d , and decisive layer ℓ_d , let

$$\Delta_i = \text{vec}(h_{i,\ell_d,\text{tail}}^D - h_{i,\ell_d,\text{tail}}^T).$$

125 We fit an uncentered SVD basis to $\{\Delta_i\}$ on training prompts, then patch only the projection of
 126 each held-out prompt’s exact donor-target difference into the learned subspace. This is a subspace-
 127 sufficiency test: at evaluation time, the held-out exact difference supplies the coordinates inside a
 128 basis learned only from training prompts. It therefore tests whether the causal mode information lies
 129 in a shared low-dimensional subspace, not whether a single fixed steering vector fully determines
 130 the branch.

131 **How the fixed vector is fitted.** The exact formula for the model- and direction-specific average is

$$\bar{\Delta}_d = \frac{1}{|S_{\text{train}}|} \sum_{i \in S_{\text{train}}} \text{vec}(h_{i,\ell_d,\text{tail}}^D - h_{i,\ell_d,\text{tail}}^T).$$

132 At held-out evaluation time, we add $\alpha \bar{\Delta}_d$ to the target activation; α is selected on training prompts
 133 only. The intervention uses the held-out target activation and the frozen training vector, but no
 134 held-out thinking-mode activation or difference. For the cross-family rows, a prompt counts as
 135 *effective* only when the modified distribution moves toward the donor by a JS margin above 0.1 and
 136 ends within 0.6 JS of it. This prevents tiny movements near the maximum possible distance from
 137 counting as successes.

138 3 The visible switch creates a short-lived internal signal

139 The first Qwen experiments test whether the public switch is serialized in the chat template and
 140 whether the resulting transition states are distinguishable from arbitrary suffix noise. On 60 prompts,
 141 the no-think suffix has a deterministic transient path: immediately before the explicit close, the
 142 model predicts `</think>` on 60/60 prompts; immediately after the close, it predicts the blank sep-
 143 arator on 60/60 prompts. The full no-think suffix then moves to direct-answer openers such as To,
 144 Let, and Here, while think mode moves to thinking-preface openers such as Thinking and Here.

145 Matched controls separate the complete template sequence from generic sensitivity to a changed
 146 suffix. On 180 Qwen prompts spanning ten categories, a real `</think>` outside the empty-block
 147 sequence is close to no-think but not identical (mean JS 0.0534). Fake close tags and moved-close
 148 controls do not reproduce the canonical mode. Same-length Okay, Think, and whitespace suffixes
 149 also remain far from thinking mode (mean JS 0.6519, 0.6767, and 0.6618). The full serialized
 150 sequence matters; merely changing the final tokens is insufficient.

151 Three further audits test simpler explanations. First, Qwen’s fixed-vector suite includes calendar,
 152 coding, dynamic programming, language, logic, and planning prompts as well as math. On the
 153 non-math categories, it reaches donor win 102/108 for no-think into think and 94/108 in reverse; on
 154 math, counting, and probability it reaches 72/72 in both directions.

155 Second, Table 4 removes likely mode-signaling tokens before computing JS. The fixed mode vec-
 156 tor remains 10/10 in both directions after removing special and whitespace tokens, common think-
 157 ing/direct openers, and a broader hand-written list. A verbose-versus-concise style vector falls to
 158 chance. With a learned mask, the no-think-into-think vector remains 9/10 after removing the 10
 159 most mode-predictive tokens, 8/10 after 100, and 9/10 after 250. Three norm-matched random
 160 vectors never exceed 3/10 and move in the wrong direction on average.

161 Third, user instructions are made deliberately misleading. Across 10 held-out prompts, seven prompt
 162 variants, and both modes, a classifier using the final activation identifies the actual template mode in
 163 140/140 cases. A heuristic using the visible instruction is correct in 60/140; a heuristic using the first
 164 predicted token is correct in 107/140. The signal is therefore not explained by an ordinary request
 165 for verbosity or a single opening token.

166 4 The causal signal is present at response entry

167 Table 5 reports exact activation-copying tests. In Qwen3.5 and Nemotron, copying the final two
 168 prompt-token activations makes every tested prompt closer to the copied mode at a tested layer, while

Table 4: Qwen token-mask audit. Each cell is held-out donor wins and JS margin after removing the listed tokens and renormalizing. The fixed mode vector survives; a visible verbosity vector does not.

Direction	Method	Unmasked	Mask openers	Mask branch markers
think into no-think	branch vector	10/10, .571	10/10, .455	10/10, .455
think into no-think	verbosity vector	7/10, .037	5/10, .044	5/10, .044
no-think into think	branch vector	10/10, .064	10/10, .108	10/10, .108
no-think into think	verbosity vector	6/10, -.001	5/10, -.000	5/10, -.000

169 copying two earlier positions never wins. Qwen3 requires a slightly different audit because the pub-
170 lic soft switch always emits a <think>...</think> wrapper. We keep enable_thinking=True,
171 use /think versus /no_think, supply the same first <think> token to both runs, and audit the next
172 decision. Copying the activation at that one shared token produces donor wins on 12/12 prompts in
173 both directions; copying the matched user-end positions four tokens earlier or the first four prompt
174 positions produces 0/12. These exact-copy tests use the matching donor run and are separate from
175 the stronger fixed-vector transfer claim. OLMo is shown separately because it compares two check-
176 points. Phi and Liquid checks remain secondary appendix results.

Table 5: Exact copying at response entry. Qwen3.5, Nemotron, and OLMo copy the final two prompt positions; Qwen3 copies the single shared <think> state at the soft switch’s first divergent decision. A successful tested boundary has donor win on every prompt in that finite suite.

Model	Donor into target	Copied width	First success	Earlier control
<i>Same-checkpoint public switch</i>				
Qwen3.5-4B	think into no-think	2	3	0.000 best donor win
Qwen3.5-4B	no-think into think	2	0	0.000 best donor win
Qwen3-4B soft	think into no-think	1	24	0.000 best donor win
Qwen3-4B soft	no-think into think	1	28	0.000 best donor win
Nemotron Nano 4B	think into off	2	16	0.000 best donor win
Nemotron Nano 4B	off into think	2	12	0.000 best donor win
<i>Paired-checkpoint comparison</i>				
OLMo 3 7B	think into instruct	2	0	0.000 best donor win
OLMo 3 7B	instruct into think	2	4	0.000 best donor win

177 After localizing the Qwen3 soft-switch decision, we fit controllers of widths 1, 2, 4, and 8 on the
178 same 12 prompts at boundary 28. Each selects the donor token on 72/72 held-out directional tests.
179 Width 1 is sufficient and has 0/432 control wins. Wider deterministic controls also have zero wins;
180 one width-8 random tensor wins on 21/36 prompts because that span reaches the switch instruction.
181 No held-out donor activation or difference enters any controller. Appendix B gives every denomina-
182 tor.

183 The module-labelled tests are asymmetric. The historical attention_only condition copies the
184 donor’s full residual state after attention, including the donor layer input, and then runs the MLP.
185 The mlp_only condition keeps the target’s post-attention state and copies only the donor MLP
186 update. In Qwen3.5-4B, the post-attention-state copy reaches donor win 1.000 in both directions;
187 the MLP-update copy reaches 0.000 in one direction and at most 0.167 in the other. At Qwen3-4B’s
188 soft-switch branch point and in Nemotron, post-attention-state copies reach 1.000 in both directions
189 while MLP-update copies remain 0.000. Thus the post-attention state reproduces the full effect
190 while the isolated MLP update does not reproduce it fully. This does not isolate the attention update
191 or one attention head.

192 5 How the signal is written and read

193 Three tests give a simple picture. First, the signal is present at response entry: exact final-prompt
194 copies work in Qwen3.5 and Nemotron, and both exact copying and held-out fixed controllers work
195 at Qwen3-4B’s shared soft-switch branch token. Second, Qwen3 width 1 succeeds on 72/72 held-
196 out tests at the branch point and 0/72 at a non-overlapping earlier position; larger widths do not

Table 6: Fixed-vector selection. The grid and rule are fixed before the held-out AIME prompt test. Effective requires JS margin > 0.1 and JS-to-donor < 0.6 .

Model	Direction	Width grid	Fixed layer	Alpha grid / selected
Qwen3.5-4B	think into no-think	{2}	15	{.25,.5,1,2,4,8} / 1
Qwen3.5-4B	no-think into think	{2}	19	{.25,.5,1,2,4,8} / 2
Gemma 4 E2B-it	think into off	{2}	35	{.25,.5,1,2,4,8} / 2
Gemma 4 E2B-it	off into think	{2}	35	{.25,.5,1,2,4,8} / 1
Nemotron Nano 4B	think into off	{2}	28	{.25,.5,1,2,4,8} / 4
Nemotron Nano 4B	off into think	{2}	28	{.25,.5,1,2,4,8} / 2

Table 7: Held-out contest-prompt test for same-checkpoint fixed vectors. Each vector is learned on 60 paired public prompts and tested on all 60 AIME 2025+2026 question strings. It uses no AIME answers, correctness labels, thinking-mode activations, or mode differences.

Model	Entry path	Think eff.	Think JS	Opp. eff.	Random eff.
Qwen3.5-4B	template suffix	60/60	0.1103	0/60	19/180
Gemma 4 E2B-it	template switch	60/60	0.0002	0/60	0/180
Nemotron Nano 4B	system prompt	60/60	0.0000	0/60	0/180

improve it. Nemotron succeeds on 42/42 broad-plus-GPQA prompts at the final two positions and 0/42 when moved 8 or 16 tokens earlier. Third, the post-attention residual copy reproduces the donor effect while the isolated MLP update does not reproduce it fully; because these interventions are asymmetric, they do not attribute the signal to attention alone.

6 Transfer beyond the training prompts

The main non-AIME test learns fixed vectors on 60 public GSM8K/MATH-500 prompts and evaluates them on instructions, coding, factual questions, multiturn conversations, non-English prompts, and GPQA Diamond. Gemma succeeds on 30/30 broad prompts (JS-to-donor 0.0204; random vectors 0/90). Nemotron succeeds on 30/30 (JS-to-donor 0.0001; random 0/90), and a reduced broad-plus-GPQA test succeeds on 42/42; moving the same vector to earlier positions gives 0/42. Qwen3-4B’s width-1 branch controller succeeds on 72/72 held-out directional tests, including 48/48 in four categories absent from fitting; widths 2, 4, and 8 do not improve this maximum. Qwen3.5’s broader width-2 fixed vector transfers less consistently; Section 11 reports that boundary.

7 Held-out contest-prompt test

We first test all 60 AIME 2025 and 2026 questions as a disjoint source of prompt text; this boundary audit does not generate or grade their answers. Width is fixed at 2 from the earlier localization test. Layer is fixed before held-out evaluation. The scale α is chosen on the 60 training prompts by prioritizing the number of effective prompts, then donor wins, JS margin, and JS-to-donor. No AIME prompt, answer, thinking-mode activation, or mode difference enters that selection. Table 7 then adds the frozen vector to each held-out off-mode activation. The later AIME 2026 generation test uses the already-frozen Nemotron controller and reports answer quality separately.

The three model-specific vectors succeed on 180/180 held-out prompt boundaries. Reversing their signs gives 0/180. Three norm-matched random vectors per model give 19/540, all in the noisier Qwen3.5 row. These are prompt-boundary mode results, not AIME accuracy results.

Secondary OLMo checkpoint-pair result. OLMo 3 7B compares separately trained Think and Instruct checkpoints. The same fixed-vector protocol reaches 60/60 held-out prompts toward Think (mean JS-to-donor 0.0207) and 60/60 toward Instruct (0.2873). This secondary result shows that the test can find an analogous activation difference across checkpoints; it is not evidence about a same-checkpoint public switch.

To summarize the leakage controls: width, layer, and the effective threshold are fixed before the held-out test; only α is selected on training prompts. AIME prompts are disjoint from training, and neither their answers nor their thinking-mode activations are used.

Table 8: Cross-validated low-rank residual subspace results. CIs are prompt-bootstrap intervals. Random subspaces of matched rank achieve donor win 0.000 in all rows.

Model	Donor into target	Rank	Donor win	Margin retention
Qwen3.5-4B	think into no-think	2	1.000 [1.000, 1.000]	0.966 [0.906, 1.046]
Qwen3.5-4B	no-think into think	2	1.000 [1.000, 1.000]	0.910 [0.869, 0.948]
OLMo 3 7B	think into instruct	2	1.000 [1.000, 1.000]	0.862 [0.811, 0.908]
OLMo 3 7B	instruct into think	1	1.000 [1.000, 1.000]	1.021 [0.963, 1.074]

8 Low-rank support

The fixed-vector audit above is the main deployable prompt-boundary test. A separate low-rank subspace analysis asks a more mechanistic question: whether the exact held-out mode difference lies in a shared low-dimensional causal subspace. Table 8 gives that result. We run 3-fold category-balanced cross-validation over a 60-prompt suite; each fold trains on 40 prompts and evaluates on 20 held-out prompts. The learned spectrum is concentrated in Qwen, where the top two components explain 0.874 and 0.854 of donor-target tail-difference energy in the two directions. OLMo has a similarly compressed paired-checkpoint subspace. The subspace interventions transfer causally; unlike the fixed-vector audit, they use the held-out prompt’s exact donor-target difference to choose coordinates inside the train-learned subspace, so they support the low-rank mechanism rather than fixed-vector deployability.

A larger Qwen-only 180-prompt subspace suite clarifies what the code represents. Full width-2 patches have donor win 1.000 with JS to donor near zero. Rank-2 SVD patches also reach donor win 1.000 in both directions; matched random rank-2 subspaces remain at donor win 0.000. Wrong-prompt exact-difference patches transfer strongly, reaching donor win 0.956 and 0.994 in the two directions. The branch code is therefore shared rather than prompt-keyed, consistent with the fixed-vector result above.

Logit attribution of the rank-2 Qwen code is branch-vocabulary-shaped. The first component in the think-donor direction promotes tokens such as *Thinking*, *thinking*, and *user-analysis* tokens, while suppressing direct-answer openers such as *To*, *To*, and *###*. This ties the residual subspace to the visible response program rather than leaving it as an abstract donor-win vector.

9 Generated behavior diagnostics

AIME 2026 answer quality after one entry decision. The frozen-before-generation protocol uses all 30 problems, one paired seed, and four conditions under the same 8,192-token budget without retries. The width-2 Nemotron off-to-thinking vector was learned only on 30 GSM8K and 30 numeric MATH-500 prompts, then frozen at boundary 28 with $\alpha = 4$. It changes one next-token decision; the token is appended to the untouched off-mode context and every later token is generated normally.

The fixed intervention enters `<think>` on 30/30 problems and scores 4/30 strictly correct, versus 1/30 for off/random and 7/30 for thinking; each paired exact test has $p = .25$. Manual review finds no false positive: fixed reaches the correct number on all seven problems solved by thinking, but fails to stop on three. Only problem 3 has a valid gold box before truncation, so removing only the completion gate gives 5/30. Fixed completes 5/30 and hits the cap on 25/30, versus 8/30 and 22/30 for thinking. Entry therefore has downstream relevance, but this one-seed result is not significant and exposes a stopping failure under a budget below NVIDIA’s 32,768-token setting (NVIDIA, 2025). Appendix F.4 gives complete tables and audits.

Other follow-through tests. On 12 disjoint Qwen3 prompts, one exact copy selects the donor newline on all 24 directional tests; after the copied state is discarded, all 72 sampled continuations follow the donor format and all 72 controls retain the target format. Across ten settings on 30 frozen non-math prompts, one Nemotron fixed-vector decision produces an exact `<think>` opening on 293/300 generations versus 0/300 for norm-matched random vectors. Reapplying that vector for 64 decisions corrupts all 30 prompts in both directions; repeated exact copies on six math prompts also

lose donor resemblance, with three of 24 modified runs repetitive at the cap. Appendix F gives the full protocols, tables, and independent answer-quality recomputation.

10 Related work

Hybrid-reasoning work studies when to spend reasoning budget and how visible triggers or steering vectors change response style (Jiang et al., 2025; Yang et al., 2026; Venhoff et al., 2025). We instead test the internal prompt-end signal written by a public switch, separating exact copies from vectors learned on other prompts. Causal-rationale work studies later reasoning traces (Rahim and Rahim, 2025; Sathyanarayanan et al., 2026); we study the state before a long trace exists. We build on activation patching and layerwise readout (Geva et al., 2023; Belrose et al., 2023; Zhang and Nanda, 2024), adding reversed, random, earlier-position, token-mask, suffix, and misleading-instruction controls.

11 Limitations

We primarily measure response entry, not faithful reasoning. The 60-row AIME boundary audit does not generate answers. The separate AIME 2026 answer test uses one paired seed over 30 problems; its bootstrap measures task, not sampling-seed, variation.

The vectors are model-specific. Qwen3.5, Gemma, and Nemotron use width-2 tails. Qwen3-4B succeeds at widths 1–8, but only at the first branch decision; width 8 reaches switch text and has a random-control win. OLMo uses separate checkpoints, and Gemma’s reverse fixed vector is unreliable.

OLMo, Phi, and Liquid compare paired checkpoints, not same-checkpoint switches; Liquid is asymmetric. Models without paired modes require a different experiment.

Transfer varies: Gemma and Nemotron succeed broadly, while Qwen3.5 reaches the strict threshold on 13/30 broad prompts versus 2/90 random controls.

Generated answers are harder because each token changes what follows. On AIME 2026, fixed enters thinking on 30/30 and scores 4/30, versus 1/30 off/random and 7/30 thinking; every paired test has $p = .25$. It reaches the correct result but fails to stop in all three losses to thinking. Fixed and thinking hit the common 8,192-token cap on 25/30 and 22/30, below NVIDIA’s 32,768-token setting, so this does not establish a general accuracy gain or recover reliable termination. Reverse entry also misses canonical direct behavior, and repeated interventions corrupt text. Reliable full-answer residual control remains unsolved.

Localization remains partial: tail post-attention residual copies work where isolated MLP updates do not, but this isolates neither attention nor one head. We find no cross-family vector or benchmark-solving policy, and proprietary switches may differ.

12 Conclusion

Public switches leave causal entry signals: Qwen3-4B controllers of widths 1–8 give 72/72 held-out donor tokens, one copy sets 72/72 continuation formats, and fixed vectors transfer in Qwen3.5, Gemma, and Nemotron. Nemotron gives 293/300 exact thinking openings and enters on 30/30 AIME 2026 problems, scoring 4/30 versus 1/30 off/random and 7/30 thinking. The differences are not significant, and its three losses to thinking are stopping failures. Entry is controllable; reasoning recovery, reliable termination, and full-answer control are not established.

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345 A Artifact map

346 The row-level artifacts are stored under `artifacts/`. Artifact directories contain run metadata plus
347 aggregate and row-level files where applicable. For double-blind review, the submitted supplement
348 should be built with `scripts/prepare_neurips_supplement.sh`; this script copies repro-
349 duction scripts, selected row-level and aggregate artifacts, prompt splits, and sanitized run metadata
350 while excluding paper source, written appendices, raw upload logs, and machine-specific shell logs.

351 The supplement also includes `scripts/verify_claim_artifacts.py`, which checks that the
352 artifact directories and primary files in the artifact map below are present, and `scripts/verify_h`
353 `eadline_numbers.py`, which checks the paper’s rounded headline values against the stored JSON
354 artifacts.

Claim	Artifact directory	Primary aggregate files
Qwen tail localization	<code>qwen35-paper-interp-20260422T075824Z</code>	<code>width_patch_aggregate.json</code>
Qwen3.5 post-attention/MLP comparison	<code>qwen35-controller-behavior-20260422T184734Z</code>	<code>module_patch_aggregate.json</code> , <code>module_patch_rows.json</code> , <code>SUMMARY.md</code>
Historical Qwen3-4B hard-template width sweep	<code>rebuttal-20260724/qwen3-4b-localization</code>	<code>patch_aggregate.json</code> , <code>first_perfect_layers.json</code> , <code>template_serialization.json</code> , <code>SUMMARY.md</code>
Qwen3-4B soft-switch prompt-tail width sweep	<code>rebuttal-20260725/qwen3-4b-soft-switch-localization</code>	<code>patch_aggregate.json</code> , <code>first_perfect_layers.json</code> , <code>SUMMARY.md</code>

Claim	Artifact directory	Primary aggregate files
Qwen3-4B public soft-switch audit	rebuttal-20260725/qwen3-4b-soft-switch-branch-point	patch_aggregate.json, module_patch_aggregate.json, first_perfect_layers.json, SUMMARY.md
Qwen3-4B same-prompt one-token diagnostic	rebuttal-20260727/qwen3-4b-soft-switch-generation	generation_aggregate.json, generation_rows.json, prompt_suite.json, SUMMARY.md
Qwen3-4B template-matched one-token replication	rebuttal-20260727/qwen3-soft-switch-one-token-followthrough	branch_rows.json, generation_aggregate.json, generation_rows.json, prompt_suite.json, SUMMARY.md
Qwen3-4B held-out fixed controller	rebuttal-20260728/qwen3-soft-switch-fixed-controller	audit_aggregate.json, audit_rows.json, alpha_selection.json, controller_diagnostics.json, SUMMARY.md
Qwen3-4B held-out fixed-controller width sweep	rebuttal-20260728/qwen3-soft-switch-fixed-controller-width-1-2-4-8	audit_aggregate.json, audit_rows.json, alpha_selection.json, token_span_audit.json, SUMMARY.md
Qwen token controls and 180 prompts	qwen35-neurips-hardening-20260425T172250Z	token_control_aggregate.json, crossval_aggregate.json
Qwen fixed steering vector	qwen35-fixed-controller-20260427T152833Z	fixed_prompt_controller_aggregate.json, fixed_prompt_controller_bootstrap_summary.json
Qwen generated-prefix rollout diagnostic	qwen35-multilayer-controller-benchmark-20260427T233149Z	generation_aggregate.json, generation_rows.json
Gated public-task steering policy	gated-controller-policy-20260505T050337Z	gated_policy_aggregate.json, gated_policy_aggregate_by_model.json, gated_policy_rows.json, crossvalidated_gated_policy_rows.json, gated_policy_paired_deltas.json
Cross-family fixed vectors	cross-family-fixed-controller-20260427T195227Z	fixed_controller_aggregate.json, fixed_controller_rows.json
Contest-style prompt-string branch-state audit	aime-fixed-branch-controller-20260429T194405Z	aime_controller_aggregate.json, aime_controller_rows.json, controller_vectors.json
Gemma/Nemotron fixed thinking-branch switch	gemma-nemotron-fixed-switch-20260505T070908Z	aime_controller_aggregate.json, aime_controller_rows.json, controller_selection.json, controller_vectors.json
Broad branch-state claim synthesis	branch-state-broad-claim-20260506T034229Z	branch_state_broad_claim_summary.json, SUMMARY.md
Branch-entry robustness audits	branch-entry-paper-audits-20260506T210000Z	paper_audit_summary.json, SUMMARY.md, svd_top_token_proxy.json
Qwen masked-JS/style control audit	qwen35-masked-js-style-control-20260506T215634Z	masked_js_style_aggregate.json, masked_js_style_rows.json, mask_metadata.json
Qwen learned branch-token mask audit	qwen35-learned-token-mask-20260507T040000Z	learned_mask_aggregate.json, learned_mask_rows.json, learned_branch_tokens_top250.json
Qwen adversarial branch audit	qwen35-adversarial-branch-audit-20260507T000000Z	classifier_aggregate.json, patch_aggregate.json, SUMMARY.md
Nemotron broad+GPQA robustness	prompt-boundary-robustness-nemotron-reduced-20260507T061500Z	fixed_aggregate.json, random_direction_summary_aggregate.json, subsets_aggregate.json, wrong_position_aggregate.json
Historical direct-erasure adapter diagnostics	shared-latent-step-direct-erasure-20260430T040633Z	shared_latent_aggregate.json, shared_latent_rows.json, latent_adapters.json
Historical Nemotron adapter stress test	shared-latent-nemotron4b-direct-erasure-step-20260505T040225Z	shared_latent_aggregate.json, shared_latent_rows.json, latent_adapters.json
Closed-loop direct-erasure diagnostic	closed-loop-shared-direct-erasure-reduced-20260505T041600Z	closed_loop_step_aggregate.json, closed_loop_prompt_aggregate.json, closed_loop_prompt_rows.json
Historical Phi kernel induction diagnostic	shared-latent-step-thinking-phi-kernel-layer28-20260430T052209Z	shared_latent_aggregate.json, shared_latent_rows.json, latent_adapters.json
OLMo localization	olmo3-localization-20260423T160830Z	block_patch_aggregate.json, module_patch_aggregate.json
Nemotron localization	nemotron-localization-20260507T000000Z	patch_aggregate.json, module_patch_aggregate.json, first_perfect_layers.json
Nemotron focused generation audit	rebuttal-20260724/nemotron-combined-analysis	combined_generation_aggregate.json, combined_generation_rows.json, fidelity_by_step.json
Nemotron one-step fixed-vector generation	rebuttal-20260724/nemotron-fixed-vector-one-step-generation	generation_aggregate.json, generation_rows.json, controller_step_rows.json, PAIRED_QUALITY_ANALYSIS.md
Nemotron 64-step stress test	rebuttal-20260724/nemotron-fixed-vector-broad-generation	generation_aggregate.json, generation_rows.json, controller_step_rows.json

Claim	Artifact directory	Primary aggregate files
Nemotron ten-seed entry audit	rebuttal-20260727/nemotron-ten-seed-entry-analysis and rebuttal-20260727/nemotron-ten-seed-entry-runs	analysis.json, SUMMARY.md, per-seed generation_rows.json and controller_step_rows.json
Nemotron AIME 2026 answer quality	rebuttal-20260728/nemotron-aime-answer-quality-protocol.json; rebuttal-20260728/nemotron-aime26-one-step-answer-quality	generation_aggregate.json, paired_answer_quality.json, independent_answer_quality_audit.json, manual_output_audit.json, MANUAL_OUTPUT_AUDIT.md
Liquid localization	liquid-causal-patch-20260426T044355Z, liquid-causal-layer-scan-20260430T051648Z	patch_aggregate.json, canonical_aggregate.json
Phi localization	phi-causal-patch-20260423T231120Z	patch_aggregate.json
Low-rank cross-validation	residual-subspace-crossval-20260424T205031Z	crossval_aggregate.json, spectrum_aggregate.json
Public behavior diagnostics	qwen35-public-benchmark-behavior-20260424T104510Z	aggregate_by_variant.json
Sustained patch generation	qwen35-sustained-patch-generation-20260425T193507Z	generation_aggregate.json, sustained_step_aggregate.json

B Qwen3 width sweeps and fixed-controller audit details

B.1 Earlier exact-copy width sweeps

Two earlier 12-prompt experiments tested widths 1, 2, 4, and 8 using the matching donor activation for each prompt. Table 10 reports them because they motivated the cleaner branch-point design, not because they establish held-out controller transfer. The first used the hard chat-template switch, `enable_thinking=True` versus `False`; its serialized prompts differ by a four-token assistant suffix. The second kept `enable_thinking=True` and used `/think` versus `/no_think`, but patched the ordinary serialized prompt tail before adding the shared `<think>` token. In that soft-switch sweep, non-tail width-2/4/8 copies also reach donor win 1.0, so it does not uniquely localize the effect to the tail. The first four prompt positions never win in either direction.

Switch and copied block	Direction	Width 1	Width 2	Width 4	Width 8
Hard template, tail	off to think	20	0	0	0
Hard template, tail	think to off	28	4	0	8
Hard template, non-tail	off to think	—	none	none	none
Hard template, non-tail	think to off	—	none	none	none
Soft instruction, tail	off to think	32	28	24	0
Soft instruction, tail	think to off	32	24	24	0
Soft instruction, non-tail	off to think	—	8	0	0
Soft instruction, non-tail	think to off	—	8	0	0

Table 10: Earlier Qwen3-4B exact-copy width sweeps. Entries are the first tested residual boundary with donor wins on all 12 prompts; the grid is 0, 4, . . . , 36, “none” means no tested boundary, and “—” means the block was not tested. Every tail width reaches 12/12 at some boundary in both directions. These are same-prompt donor copies, not fixed controllers, and have no fit/held-out split.

B.2 Held-out fixed-controller width sweep

The new width protocol, runner, prompt IDs, layer boundary, alpha grid, selection rule, controls, random seeds, and interpretation limits were committed before inference. Both conditions use model revision 4dcb3d101c2a062e5c1d4bb173588c54ea6c4d25, keep `enable_thinking=True`, and differ only through the public `/think` and `/no_think` instructions. We supply the same `<think>` token, then audit the first divergent next-token decision. For each width, the frozen controller is the mean thinking-minus-off residual tensor over the same 12 fit prompts at boundary 28. The fit-only selection rule chooses $\alpha = 1$ for every width. Evaluation uses 12 new values from the same six authored categories and 24 prompts from four categories absent from fitting. It never uses a held-out donor activation or difference.

Width 1 already reaches the maximum binary result, so more positions add no success benefit. Its mean JS-to-donor is 8.05×10^{-9} off-to-think and 9.94×10^{-7} think-to-off; mean donor margins are 0.69315 and 0.69313. Controller norms for widths 1/2/4/8 are 112.587/144.613/201.942/303.778, and the corresponding mean fit-prompt cosines to the mean tensor are 0.975/0.951/0.963/0.960. The

Width	α	Fixed wins	Token matches	Zero	Reversed	Earlier span	Random
1	1	72/72	72/72	0/72	0/72	0/72	0/216
2	1	72/72	72/72	0/72	0/72	0/72	0/216
4	1	72/72	72/72	0/72	0/72	0/72	0/216
8	1	72/72	72/72	0/72	0/72	0/72	21/216

Table 11: Held-out Qwen3-4B fixed-controller width sweep over 36 prompts and both directions. Every fixed row comprises 24 same-category and 48 category-shift directional tests. The earlier-span control has the same width but never overlaps the active response-entry span. Random denominators contain three preregistered norm-matched tensors. All 21 random wins are one width-8 seed in the off-to-think direction.

tokenizer audit shows that widths 1, 2, and 4 stay within the shared `<think>` and assistant-header suffix, while width 8 reaches the differently tokenized `/think` and `/no_think` instruction fragments. This explains why width 1 is the cleanest sufficiency and specificity result. These tests do not measure accuracy, reasoning quality, or later-token control.

C Localization effect sizes

Table 12 expands the headline localization rows from Table 5. The Qwen3.5 JS columns are point estimates from the stored aggregate. Qwen3, Nemotron, OLMo, Liquid, and Phi include prompt-bootstrap intervals. Qwen3 reports the shared-token soft-switch decision after both modes receive `<think>`. Phi reports the main 8-prompt set; the alternate set reaches the same first successful tested boundaries.

Model	Donor into target	n	Layer	Donor win	JS to donor	JS to target
Qwen3.5-4B	think into no-think	60	3	1.000 [1.000, 1.000]	0.0632	0.6283
Qwen3.5-4B	no-think into think	60	0	1.000 [1.000, 1.000]	0.1113	0.6080
Qwen3-4B soft	think into no-think	12	24	1.000 [1.000, 1.000]	0.0000001	0.6931
Qwen3-4B soft	no-think into think	12	28	1.000 [1.000, 1.000]	0.0000002	0.6931
Nemotron Nano 4B	think into off	12	16	1.000 [1.000, 1.000]	0.1404	0.6893
Nemotron Nano 4B	off into think	12	12	1.000 [1.000, 1.000]	0.1222	0.6748
OLMo 3 7B	think into instruct	12	0	1.000 [1.000, 1.000]	0.2552	0.6925
OLMo 3 7B	instruct into think	12	4	1.000 [1.000, 1.000]	0.5022	0.6927
Liquid LFM2.5 1.2B	think into instruct	60	16 best	0.000 [0.000, 0.000]	0.6924	0.6037
Liquid LFM2.5 1.2B	instruct into think	60	12	1.000 [1.000, 1.000]	0.1246	0.6931
Phi-4 mini	think into instruct	8	28	1.000 [1.000, 1.000]	0.0110	0.6932
Phi-4 mini	instruct into think	8	16	1.000 [1.000, 1.000]	0.2996	0.6921

Table 12: Effect sizes for the response-entry localization rows. Qwen3 uses a width-1 copy of the shared `<think>` state; the other headline rows use width 2. Qwen3, Nemotron, and OLMo boundaries are the first successes on coarse four-layer grids, not exact onsets. The Liquid think-into-instruct row reports the best tested tail boundary because no tested boundary produced donor wins. Matched non-tail controls have donor win 0.000 [0.000, 0.000] for all rows with bootstrap intervals in the artifacts.

D Compute and existing assets

Experiments were run on an Apple M4 Mac Mini with 16GB unified memory using MLX and MLX-LM. The BF16 Qwen replication covers fewer configurations than the 4-bit cross-validation because of this memory limit. The sustained-generation suite is also smaller because it recomputes full donor and target traces at each generated step. The supplement’s per-run `run_metadata.json` files record UTC start times, platform strings, model identifiers, layer and alpha grids, prompt counts, and generation caps for the included runs.

The AIME 2026 answer-quality experiment contains 120 generations under one equal 8,192-token cap frozen before generation. We report every cap hit without retrying it. This budget is one quarter of the 32,768-token setting in NVIDIA’s public model-card example (NVIDIA, 2025), so the experiment directly compares the four conditions at equal cost but does not measure their accuracy under NVIDIA’s longer setting.

The model checkpoints are open Hugging Face assets: Qwen3.5-4B and Qwen3-4B are Apache-2.0, OLMo 3 7B Think/Instruct is Apache-2.0 with Ai2 Responsible Use Guidelines, Gemma 4 E2B-it is documented by Google DeepMind with thinking-mode controls, LiquidAI LFM2.5 1.2B uses the LFM Open License v1.0, Phi-4-mini-reasoning is MIT, and Llama-3.1-Nemotron-Nano-4B-v1.1 uses the NVIDIA Open Model License with additional Llama 3.1 terms. The prompt-boundary branch audits use math-ai/aime25 and math-ai/aime26, which are Apache-2.0 licensed on their Hugging Face dataset pages. Public generation diagnostics are provided as behavior artifacts for the same intervention family.

E Broader impacts

This work gives a concrete audit target for hybrid-reasoning systems: an activation measured before generation is easier to test than a vague claim that thinking is on. The main dual-use risk is superficial manipulation of thinking-style behavior, so we pair the result with held-out tests and failures that distinguish mode entry from answer quality.

F Generated behavior diagnostics

F.1 Qwen3 template-matched one-token replication

The follow-through suite freezes every choice made by the 12-prompt localization audit, then uses the next two authored instances from each of its six categories and templates. The resulting 12 prompts are disjoint from localization but category- and template-matched. This tests new prompt values within the same authored structures, not broad-distribution transfer. Both modes keep `enable_thinking=True`, use `/think` or `/no_think`, and receive the same supplied `<think>` token. The copied residual state is used only to choose the next argmax token at layer 24 for off-to-think and layer 28 for think-to-off. We then delete every copied state, insert the selected token into the untouched target context, and sample with temperature 0.6, top-p 0.95, and top-k 20. Each prompt and direction uses three seeds with common random numbers across conditions. The design reports 216 direction-labeled analysis rows. Because a canonical reference completion serves as the donor in one direction and the target in the other, these rows correspond to 144 prompt/context/seed generation conditions.

Table 13: Qwen3-4B response format after one residual-selected branch token on 12 disjoint, template-matched prompts under three sampling seeds. The target control inserts the unmodified target mode’s argmax token into the same target context.

Direction	Condition	Observed format	Completed	Reopened think	Mean tokens
Off to think	Donor-token reference	36/36 thinking	36/36	0/36	1976.9
Off to think	One-token mediation	36/36 thinking	36/36	0/36	1809.1
Off to think	Target-token control	36/36 direct	36/36	0/36	224.2
Think to off	Donor-token reference	36/36 direct	36/36	0/36	224.2
Think to off	One-token mediation	36/36 direct	36/36	0/36	218.1
Think to off	Target-token control	36/36 thinking	36/36	0/36	1976.9

The copied state selects the donor token on 12/12 prompts in each direction. The 72 mediated continuations then match the donor response format, while all 72 paired target-control rows retain the target format. Every row has exactly one closing think tag and a nonempty answer, and none reaches the 4,096-token limit. The intervention also transfers the large length difference: off-to-think averages 1809.1 tokens versus 224.2 for its direct control, while think-to-off averages 218.1 versus 1976.9 for its thinking control. This establishes that the causally selected branch decision is sufficient for the later visible format on this template-matched prompt-value replication. It does not show broad-distribution transfer or that the discarded residual state controls later tokens.

F.2 Earlier Qwen generated-prefix diagnostics

Generated-prefix tests ask what happens after sampled tokens begin changing the next model state. In a Qwen diagnostic, a vector is learned on 24 public prompts and evaluated on 40 disjoint

GSM8K/MATH-500 prompts. It is added for the first generated decision and then for up to 64 later decisions before the model continues freely. The selected no-think-to-think vector changes response style and performs favorably on this one held-out set; the reverse row shows that style movement need not improve answers.

	Condition	Acc.	Direct	Self-corr.	Tokens	Trunc.
	canonical no-think	0.725	0.925	0.075	530.5	0.150
	canonical think	0.200	0.125	0.850	1012.8	0.950
	no-think → think generated-prefix vector	0.800	0.000	0.000	516.1	0.100
	think → no-think generated-prefix vector	0.075	1.000	0.225	1013.0	0.950

The serialized-suffix diagnostic asks whether downstream behavior moves after the prompt-boundary state enters the transient no-think branch. On a separate 24-prompt public subset, canonical Qwen think is much longer and more self-correcting but truncates on 75.0% of prompts; no-think transient states are shorter, less self-correcting, and match no-think-style direct-answer behavior. Accuracy is included to confirm complete responses.

	Variant	Think	Direct	Self-corr.	Acc.	Tokens	Trunc.
	think suffix	0.750	0.167	0.917	0.417	3585.9	0.750
	shared prefix	0.208	0.708	0.500	0.500	2022.2	0.375
	no-think preclose	0.000	0.875	0.167	0.708	984.4	0.125
	closed, no gap	0.000	0.875	0.167	0.708	1028.0	0.125
	full no-think suffix	0.000	0.875	0.250	0.708	1071.7	0.125

F.3 Nemotron generation details

Ten-seed entry-format audit. We first test the opening under ten paired generation/random-vector seed settings, using the same 30 frozen non-math prompts and the same frozen width-2 controller. Each run changes exactly the first generated decision and is capped at 64 tokens. We pre-specify an exact <think> opening as the forward generated-format endpoint. The fixed vector reaches that endpoint on 293/300 generations, versus 0/300 for the norm-matched random vectors. The paired difference is .977. A two-way bootstrap that resamples both prompts and paired run settings gives a 95% interval of [.930, 1.000] from 10,000 resamples. Every individual run scores between 27/30 and 30/30 for the fixed vector, while every random-vector run scores 0/30.

Table 14: Nemotron ten-seed branch-entry audit. “First-step donor win” compares the intervened next-token distribution with the canonical donor and target distributions; the exact-opening row instead evaluates sampled text.

Endpoint	Fixed vector	Random vector	Per-run counts
Exact <think> opening, off to think	293/300	0/300	Fixed 27–30/30; random 0/30
First-step donor win, off to think	300/300	5/300	Fixed 30/30; random 0–2/30
First-step donor win, think to off	300/300	0/300	Fixed 30/30; random 0/30

All 300 fixed off-to-think interventions select <th as the first token. The seven strict opening failures occur during ordinary sampling after that intervened token and produce malformed continuations such as <think offhand or <think off-topic>; they remain counted as failures. They are concentrated on three prompt values: multiturn_02 twice, nonenglish_02 once, and nonenglish_04 four times. We omit a reverse generated-format rate because merely failing to open with <think> does not distinguish a clean direct answer from an unrelated or malformed random-vector output. We instead report the saved first-step donor-win endpoint in both directions. For each prompt, the fixed first-step distribution is deterministic and therefore repeats across the ten run settings; the 300/300 fixed rows in each direction are descriptive, not 300 independent distributional replications. The random-vector controls vary across runs. Reverse fixed generation is also greedy, so its output repeats across seeds. The 64-token cap makes this an entry-format audit; the separate experiment below supplies the 1,024-token completion, length, and repetition evidence.

F.4 AIME 2026 answer quality after a frozen entry intervention

This experiment asks the reviewer’s downstream question directly: what answer quality follows after the recovered thinking branch is entered? The controller is the previously frozen Nemotron

off-to-thinking vector: width 2, boundary 28, and $\alpha = 4$, fitted and selected only on the 30 GSM8K and 30 numeric MATH-500 prompts. The operational protocol was frozen before any AIME 2026 generation. It uses all 30 AIME 2026 problems, one paired seed (20260728), four conditions, and one equal 8,192-token budget without retries. The shared cap was frozen as an equal resource budget, but it is one quarter of the 32,768 tokens in NVIDIA’s model-card generation example (NVIDIA, 2025). An earlier AIME 2025 timing pilot is excluded from every result.

Canonical off generation is greedy; the other three conditions use the protocol’s thinking sampler. For the fixed and norm-matched random conditions, the intervention changes only the first next-token distribution. The sampled token is appended to the untouched off-mode prompt, the patched state is discarded, and the model generates every later token normally. The primary score requires the response to stop before the cap and its last syntactically valid `\boxed{N}` integer to match the gold answer. A final standalone-integer parser is secondary. Table 15 reports every condition; no entry, completion, cap, or repetition failure is removed.

Table 15: Nemotron pass@1 answer quality on all 30 AIME 2026 problems. “Entry” is an exact initial `<think>` tag; capped responses are incomplete and incorrect under the strict metric.

Condition	Entry	Strict	Complete	Cap	Tail rep.	Mean tokens
Canonical off	0/30	1/30	20/30	10/30	13/30	3449.7
Canonical thinking	30/30	7/30	8/30	22/30	0/30	7408.9
Fixed off-to-thinking entry	30/30	4/30	5/30	25/30	0/30	7551.1
Norm-matched random entry	0/30	1/30	22/30	8/30	9/30	4042.1

Table 16: Paired strict-answer comparisons for the fixed one-step entry condition on AIME 2026. Intervals are 10,000-resample task-bootstrap intervals; p is the exact two-sided McNemar test over discordant problems.

Reference	Fixed / reference	Difference [95% CI]	W/L/T	p
Canonical off	4/30 / 1/30	+ .100 [.000, .233]	3/0/27	.25
Canonical thinking	4/30 / 7/30	− .100 [−.233, .000]	0/3/27	.25
Norm-matched random	4/30 / 1/30	+ .100 [.000, .233]	3/0/27	.25

The fixed vector produces the exact `<think>` opening, the thinking donor’s top-1 entry token, a donor win, and effective control on all 30 problems. Its successful-entry subset is therefore the full fixed-condition population. The observed 4/30 accuracy is above off and random but below canonical thinking; none of the differences is significant. Moreover, 25/30 fixed responses hit the shared cap and only 5/30 complete.

Manual review found no strict false positive. Fixed reaches the correct numerical result on all seven problems that canonical thinking scores strictly correct. It receives strict credit on four; on problems 3, 4, and 8 it reaches the same correct result but continues reasoning until the cap instead of stopping. Problem 3 writes a valid gold `\boxed{79}` before closing the think block, restarting, and truncating; dropping only the completion gate would make fixed 5/30, but no other capped response contains a valid gold box. Seven capped traces reach or discuss the gold result (six fixed and one canonical thinking), but we do not score post-hoc number mentions because unfinished traces also consider and retract candidate values. The fixed-versus-thinking strict gap therefore reflects a termination failure on every discordant problem, while the strict 4/30 remains the primary frozen score.

The high cap rates—25/30 fixed and 22/30 canonical thinking—limit the answer-quality comparison under an 8,192-token budget. The experiment supports a causal link from the recovered state to branch entry and shows that the entered branch can carry nonzero answer quality, but it does not establish recovery of canonical thinking, reliable stopping, or performance under the longer public model-card setting.

An independent post-run script reconstructs the frozen prompts and vectors, verifies the recorded token IDs against decoded text, recomputes strict and secondary correctness, checks the cap-hit subset, and regenerates every aggregate, bootstrap interval, paired count, and McNemar test. It passes on all 120 unique prompt-by-condition rows; the hash manifest, machine-readable report, completed manual audit, and its navigation queue are included with the artifact.

511 **One fixed-vector step on 30 held-out non-math prompts.** We use the frozen width-2 vector
 512 learned from the 60 GSM8K/MATH-500 training prompts and apply it for exactly the first generated
 513 decision. We then let the model continue freely for up to 1,024 tokens. The evaluation set has
 514 30 previously frozen prompts: six each from five non-mathematical categories. The experiment
 515 contains exactly 180 generations: two canonical modes, two fixed-vector directions, and two norm-
 516 matched random-vector directions for every prompt.

Table 17: Nemotron generation after one fixed-vector step. “Donor opening” means starting with `<think>` in the forward direction and starting without it in reverse. The random vector is norm matched and uses the same scale and decoding policy.

Condition	Donor opening	Completed	Repetitive	Mean tokens
Canonical off	–	24/30	0/30	643.7
Canonical thinking	–	16/30	0/30	889.1
Off plus thinking vector	29/30	15/30	0/30	863.1
Thinking plus off vector	30/30	12/30	0/30	857.0
Random vector, forward	0/30	24/30	0/30	697.5
Random vector, reverse	0/30	12/30	0/30	915.7

517 The fixed and random vectors differ on every prompt at the first generated decision in both directions.
 518 The fixed vector produces the donor opening on 29/30 forward prompts versus 0/30 for random
 519 (paired exact $p = 3.73 \times 10^{-9}$) and 30/30 reverse prompts versus 0/30 for random ($p = 1.86 \times 10^{-9}$).
 520 None of the 60 fixed-vector outputs triggers the pre-specified repetition test, and none has low token
 521 diversity over its first 64 tokens. The one forward miss begins with the malformed fragment `<think`
 522 `offhand`; two reverse outputs enter a think block later.

523 The forward effect continues beyond the opening: its completion count is 15/30 versus 16/30 for
 524 canonical thinking, and its mean length differs by -26.0 tokens (95% prompt-bootstrap interval
 525 $[-78.8, 25.1]$). The reverse intervention selects the direct opening but does not restore the full
 526 direct-answer trajectory: it completes 12/30 responses versus 24/30 for canonical off ($p = .00183$)
 527 and is 213.3 tokens longer on average (95% interval $[124.5, 308.5]$). This is reliable entry control,
 528 not symmetric control of the complete response.

529 As a stress test, we reapplied the same fixed vector for 64 generated decisions on the same 30
 530 prompts. The first-step records and canonical baselines are identical to the one-step run, but all 30
 531 forward outputs begin with a repeated `<th` fragment and all 30 fixed-vector outputs in each direction
 532 fail the carried-forward first-64 diversity test. Nine of 30 reverse outputs also meet the full-output
 533 repetition rule. The corruption is caused by repeatedly forcing a prompt-end vector after generation
 534 has moved into a different state; it is evidence against using this fixed intervention as a long-horizon
 535 controller.

536 **Repeated exact activation copies on six math prompts.** The second test uses six public math
 537 prompts. At each of the first 64 generated decisions, it recomputes the donor and target trajectories
 538 using their respective mode prompts plus the same generated prefix, then replaces the target’s final
 539 two residual positions with the matching donor positions. After 64 such decisions it continues freely.
 540 Thinking mode used three seeds per prompt, giving 18 thinking runs and 18 off-runs modified with
 541 the matching thinking activation. The reverse direction used one run per prompt. Table 18 includes
 542 every run, including failures.

Table 18: Nemotron generation after recomputing and reapplying the matching donor tail activation at each of the first 64 decisions, then continuing freely. “Correct (all)” retains the last reported answer even for a repetitive run; “correct (finished)” excludes runs still repeating at 8,192 tokens.

Condition	Finished	Degenerate	Correct (all)	Correct (finished)
Unmodified off mode	6/6	0	4/6	4/6
Unmodified thinking mode	18/18	0	17/18	17/18
Off run given thinking activation	17/18	1	17/18	17/17
Thinking run given off activation	4/6	2	6/6	4/4

543 The intervention is strongest at response entry and then fades even while the matching copy is reap-
 544 plied. For off runs given the thinking activation, cumulative donor-win rates across the first 1, 4, 8,

16, 32, and 64 patched decisions are 1.000, .958, .694, .535, .450, and .382. In the reverse direction they are 1.000, .708, .542, .417, .349, and .284. Manual inspection found two reverse runs repeating 477 and 350 boxed answers and one forward run cycling among incompatible interpretations. All five initially capped runs were repeated with an 8,192-token limit and exactly reproduced their original 4,096-token prefixes. Complete modified answers often match the donor mode, but these failures show that repeated exact copying is not reliable long-horizon control.

G Plain-language glossary

Term	Intuitive meaning	Exact use in this paper
Thinking switch	The public setting that asks a model to produce a thinking-style response	A chat-template option for Qwen and Gemma; a system instruction for Nemotron
Activation	The model’s internal state at one token and layer	A vector read from the residual stream before the next transformer layer
Prompt end or tail	The point immediately before the model writes its answer	The final one, two, four, or eight input-token positions; width 2 is the main setting
Donor and target	The mode copied from, and the run being changed	The donor activation replaces or shifts the target activation for the same user prompt
Exact patch	Copy the actual internal state from the matching donor run	Used to locate where and through which module the mode signal is carried
Fixed mode vector	One reusable average shift learned on other prompts	Mean paired donor-minus-target activation, scaled on training prompts only; 60 fit prompts in the original suites and 12 in the Qwen3 branch-point audit
Donor win	The changed run looks more like the desired mode than its original mode	Its next-token distribution has lower JS distance to the donor than to the target
Effective	A donor win large enough to matter under a fixed threshold	Donor win with JS margin > 0.1 and JS-to-donor < 0.6
Post-attention state / MLP update	The full residual state after attention, versus the MLP’s added update	The former is copied in full; the latter is copied alone, so the asymmetric result does not isolate attention
Low-rank subspace	A small set of shared activation directions	An SVD basis fitted on training differences; unlike the fixed vector, evaluation uses each held-out donor difference’s coordinates

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